



Scan2BIM MVP Infrastructure and Cost Sheet

Purpose: This page defines the recommended **MVP infrastructure** for the Scan2BIM project and provides a practical **cost sheet** for budget approval, implementation planning, and future scaling.

The MVP should support the full Scan2BIM pipeline end to end: ingesting large E57 scan files, orchestrating a multi-step workflow, running GPU-based segmentation, executing CPU-heavy post-processing, and delivering IFC outputs back to the external platform. The design below intentionally balances **low-to-moderate cost** with a clean upgrade path to production.

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Executive summary

The recommended MVP is an **Azure-first deployment** built around **AKS, Azure Blob Storage, Azure Service Bus, MongoDB Atlas**, and a **self-hosted Temporal workflow engine**. The only cross-cloud dependency is **AWS Cognito**, which is required to authenticate with the external DataRepo integration.

- The MVP is designed for **2-5 scans per day**.
- Expected scan processing time is **45-120 minutes**, depending on scan size and GPU availability.
- The biggest cost driver is the **GPU node**; using **spot pricing** and **scale-to-zero** keeps costs under control.
- Expected MVP operating cost: **~\$1,100-\$1,600/month**.

Recommendation: Proceed with the MVP architecture below. It is cost-efficient, technically viable for early delivery, and can be scaled to production without redesigning the platform.

Recommended MVP infrastructure

Layer	Recommended service	Purpose	MVP decision
Container platform	Azure Kubernetes Service (AKS)	Runs Root MS, Segmentation MS, Post-Processing MS, and platform components	Use a small AKS cluster with one standard node and one GPU node pool
GPU compute	Azure NC6s_v3 Spot VM in AKS node pool	Runs AI segmentation workloads	Use spot capacity with KEDA scale-to-zero
Messaging	Azure Service Bus Standard	Decouples services and supports retries	Three queues plus DLQ handling
Object storage	Azure Blob Storage Hot LRS	Stores intermediate scan artifacts and IFC backups	Start with 500 GB
Workflow state	Temporal self-hosted	Tracks long-running workflows and retries	Minimal deployment for MVP
Workflow database	Azure Database for PostgreSQL Flexible Server	Backend persistence for Temporal	Burstable B2s is sufficient for MVP
Application database	MongoDB Atlas M10	Stores workflow metadata, execution state, deduplication, and logs	Shared tier acceptable for MVP
Image registry	Azure Container Registry Basic	Stores deployable	Basic tier is sufficient

		service images	
Secrets	Azure Key Vault	Manages sensitive configuration and decryption keys	Use with SOPS and workload identity
Ingress	NGINX Ingress + Azure DNS + cert-manager	Exposes webhook/API endpoints securely	TLS enabled, public entry only where required

MVP architecture scope

The MVP includes the following runtime components:

- **Root Microservice** – receives webhook events, creates workflows, coordinates the pipeline
- **Segmentation Microservice** – runs GPU-heavy AI inference on E57 data
- **Post-Processing Microservice** – performs geometry extraction, IFC generation, and final delivery
- **Temporal** – manages orchestration, state, and retries
- **KEDA** – scales segmentation workers based on queue depth
- **Flux CD** – supports GitOps deployment model

📌 **MVP trade-off:** The design favors **cost efficiency over redundancy**. This is appropriate for a first release, but it means reduced availability, slower recovery, and a higher dependency on operational monitoring.

Logical flow

1. User uploads an E57 scan through the external portal.
2. Portal/DataRepo triggers the Root service through a webhook.
3. Root service creates a workflow and pushes a message to the segmentation queue.
4. KEDA scales the GPU worker when a segmentation message appears.
5. Segmentation downloads the scan, processes it, and writes PLY/intermediate files to Blob Storage.

6. Post-Processing consumes the next message, downloads outputs, extracts geometry, and builds the IFC file.
7. Final IFC is uploaded to DataRepo using Cognito-authenticated access and also stored in Blob as backup.
8. Workflow is marked complete in Temporal and MongoDB.

Infrastructure sizing for MVP

Component	SKU / Size	Planned usage	Why this is enough for MVP
AKS standard node	1 × Standard_D4s_v3	Always on	Can host Root MS, Post-Processing MS, Temporal minimal stack, ingress, and platform utilities for low traffic
AKS GPU node pool	1 × Standard_NC6s_v3 Spot	On-demand via autoscaling	Enough for one active segmentation workload at a time
Service Bus	Standard	3 queues + DLQs	Message sizes are small, so Premium is not required yet
Blob Storage	Hot LRS, 500 GB	Intermediate artifacts + output backup	Sufficient for early scan volumes if lifecycle retention is managed
MongoDB Atlas	M10 Shared	Metadata and execution state	State data is light relative to scan file storage
PostgreSQL	Burstable B2s	Temporal persistence	Adequate for low workflow concurrency
ACR	Basic	Container image storage	Low image volume in MVP

Cost sheet

Cost item	Service / SKU	Estimated monthly cost	Cost behavior	Notes
AKS standard compute	1 × D4s_v3	\$140	Fixed	Baseline cluster compute
GPU compute	1 × NC6s_v3 Spot	\$200-\$600	Variable	Depends on scans per day and queue activity
Azure Service Bus	Standard tier	\$10	Fixed	Three queues included
Azure Blob Storage	Hot LRS, 500 GB	\$10-\$15	Variable	Includes storage plus transactions
MongoDB Atlas	M10	\$60	Fixed	Shared tier
Azure Container Registry	Basic	\$5	Fixed	Image storage for deployments
PostgreSQL for Temporal	Burstable B2s	\$25	Fixed	Minimal Temporal backend
Azure Key Vault	Standard	\$3-\$5	Fixed	Secrets and SOPS key handling
Azure DNS	Hosted zone	\$1	Fixed	Minimal public DNS cost
Network egress / buffer	Azure bandwidth + overhead	\$100-\$200	Variable	Budget buffer for transfers and unexpected usage
Total MVP	Estimated range	\$554-\$1,061 core + GPU variability	Mixed	Practical planning range:

				~\$1,100-\$1,600/ month
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GPU cost sensitivity

Usage scenario	Approximate GPU usage	Estimated GPU monthly cost
Low activity	0 scans/day	\$0
Moderate MVP load	5 scans/day at ~30 min each	~\$330/month
Upper MVP load	10 scans/day	~\$660/month

Key risks and mitigations

Risk	Impact	MVP mitigation	Status
Spot GPU preemption	Interrupted segmentation job	Use Temporal + queue retry; accept added delay	ACCEPTED
Single standard node failure	Temporary service outage	Keep backup manifests and rapid redeploy process; add second node in next phase	ACCEPTED
Storage growth	Higher monthly cost and slower cleanup	Apply lifecycle policy and retention rules for intermediate files	MONITOR
Workflow/database bottlenecks	Slower orchestration at higher concurrency	Upgrade PostgreSQL and MongoDB when volume increases	MONITOR
Integration token/auth failures	Failed DataRepo delivery	Implement token refresh, retry, and alerting	PLANNED

Implementation plan

- ☐ Create Azure resource group and networking baseline
- ☐ Provision AKS with one standard node and one GPU spot node pool
- ☐ Install NGINX Ingress, cert-manager, KEDA, and Flux CD
- ☐ Create Service Bus, Blob Storage, ACR, Key Vault, PostgreSQL, and MongoDB Atlas
- ☐ Deploy Temporal minimal stack
- ☐ Deploy Root, Segmentation, and Post-Processing services
- ☐ Configure DataRepo webhook and Cognito credentials
- ☐ Validate full upload-to-IFC pipeline with sample scans